ModelIt!

Student Activity Pack

What Happens When Habitats Change?

How Animals Survive Environmental Change

Grade 3 | NGSS 3-LS4-4 | Cause and Effect

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What would happen to a fish if its pond dried up? What choices would the fish have?

2 Have you ever seen a place where trees were cut down or a fire burned the land? What happened to the animals?

3 Draw a habitat before and after a big change, and show what happens to the animals that lived there.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Habitat Change | Food Availability | Animal Population

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Small Change

Habitat Change set to low

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Major Change

Habitat Change set to maximum

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Moderate Change

Habitat Change set to medium

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Shelter Availability — How much shelter and safe hiding places remain after the habitat changes — animals need cover from predators and weather

Species Flexibility — How well an animal can switch to different foods, shelters, or behaviors when its habitat changes

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

What happens to animals when their habitat suddenly changes — do they all survive, or do some struggle? Use evidence from your simulation to support your answers.

1 Why are animals that eat only one type of food at greater risk when habitats change than animals that eat many foods?

2 How did your model help you understand why some animals survive habitat changes while others do not?

8 STEM Challenge: Design a Wildlife Rescue Plan

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YOUR ENGINEERING MISSION

Create a plan to help animals survive after a major habitat change, using strategies like wildlife corridors, food stations, or habitat restoration.

THE SCENARIO

A forest fire has destroyed half of a national park's habitat. The park rangers need your help creating a plan to save the animals that live there. Your team must design solutions using what you learned about habitat change and animal survival!

Engineering Design Thinking

1. WHAT WILDLIFE CORRIDORS OR SAFE PATHS WILL YOU CREATE FOR ANIMALS TO MOVE TO HEALTHY HABITAT?

2. WILL YOU SET UP TEMPORARY FOOD OR WATER STATIONS FOR ANIMALS IN NEED?

3. HOW WILL YOU HELP THE FOREST REGROW — PLANTING SEEDS, PROTECTING YOUNG TREES?

4. HOW WILL YOU KNOW IF YOUR PLAN IS WORKING — WHAT SIGNS OF RECOVERY WILL YOU LOOK FOR?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! **Explain your design using scientific evidence from the simulation:**

Real Career Connection

Conservation Biologists work to protect endangered species and restore damaged habitats around the world. They earn $45,000-$90,000/year.